

## Section & Detail Views Guide

A student reference for showing internal features and enlarged details clearly in engineering drawings.

### Use this guide when...

- hidden lines make a view crowded or difficult to read.
- you need to show an internal feature, wall thickness, hole, pocket, or slot.
- a small feature needs to be enlarged for communication or dimensioning.

## What each view does

| View type      | Purpose                                                                 | Use it when                                                               |
|----------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Section view   | Shows what the object looks like after it is cut by an imaginary plane. | Internal geometry is important or hidden lines are too cluttered.         |
| Full section   | Cuts completely through the object along one plane.                     | The internal shape is symmetric or easy to explain with one straight cut. |
| Half section   | Shows half exterior and half interior.                                  | The object is symmetric and both inside/outside information matters.      |
| Offset section | Bends the cutting plane to pass through multiple features.              | Important features are not lined up on one straight plane.                |
| Detail view    | Enlarges a small area of the drawing.                                   | A small feature cannot be read or dimensioned at the main view scale.     |

## Section view basics

- A cutting plane line shows where the object is sliced.
- Arrows on the cutting plane show the viewing direction.
- The section view label matches the cutting plane label, such as A-A.
- Hatch lines identify solid material cut by the plane.
- Empty spaces such as holes, pockets, and voids are not hatched.

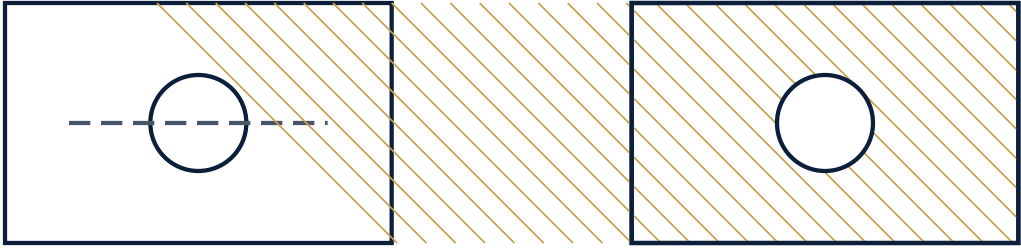
# Full section example

The cutting plane identifies where the part is sliced; the section view shows the cut interior clearly.

Cutting plane A-A

→ — — — — ←

Section A-A



Original view

Section view

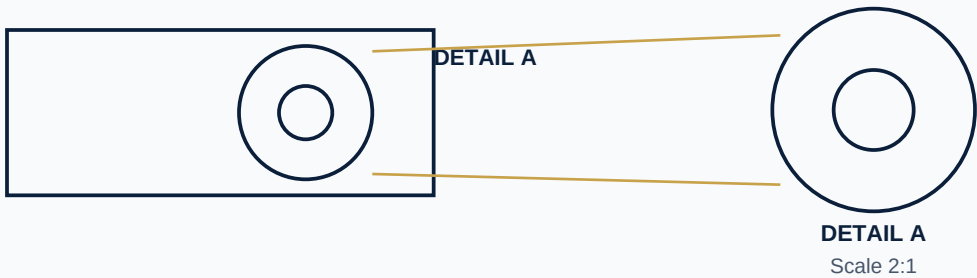
Hatch only the solid material that the cutting plane passes through.  
The hole/void remains blank because empty space is not material.

## Hatching rules

| Rule                                                                   | Why it matters                                                           |
|------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Hatch only solid cut material.                                         | Hatching a void or hole incorrectly suggests that material exists there. |
| Keep hatch spacing and angle consistent within one part.               | Consistent hatching makes the section view easier to read.               |
| Do not hatch thin features so heavily that they become dark blocks.    | The reader still needs to see edges, holes, and dimensions.              |
| Avoid hidden lines in section views unless they add important clarity. | The section view usually replaces hidden-line clutter.                   |
| Label the section clearly, such as SECTION A-A.                        | The reader must know which cutting plane produced the section view.      |

# Detail view example

A detail view enlarges a small feature so it can be read or dimensioned clearly.



## Detail view rules

| Rule                                                                 | Why it matters                                                  |
|----------------------------------------------------------------------|-----------------------------------------------------------------|
| Circle or otherwise identify the area being enlarged.                | The reader needs to know where the detail came from.            |
| Label the enlarged view, such as DETAIL A.                           | The detail label connects the enlarged view to the source area. |
| Include the detail scale when needed.                                | The enlarged detail may not be the same scale as the main view. |
| Use a detail view only for a feature that benefits from enlargement. | Too many detail views can make a drawing harder to use.         |

## Common mistakes to avoid

| Mistake                                                  | Better choice                                                      |
|----------------------------------------------------------|--------------------------------------------------------------------|
| Hatching a hole, pocket, or empty space.                 | Leave voids blank; hatch only solid cut material.                  |
| Forgetting arrows on the cutting plane.                  | Add arrows so the viewer knows the viewing direction.              |
| Using a section view without a matching label.           | Match the cutting plane and view labels, such as A-A.              |
| Using hidden lines everywhere instead of a section view. | Use a section view when it communicates the interior more clearly. |
| Adding a detail view without scale or source area.       | Show where the detail comes from and label the enlarged view.      |

## Quality checklist

- The section or detail view has a clear reason for being included.
- The cutting plane line, arrows, and section label match.
- Hatch lines appear only on solid material cut by the plane.
- Holes, pockets, and empty spaces are not hatched.
- The section/detail view is easier to interpret than a hidden-line view.
- Detail views identify the source area and include scale when needed.